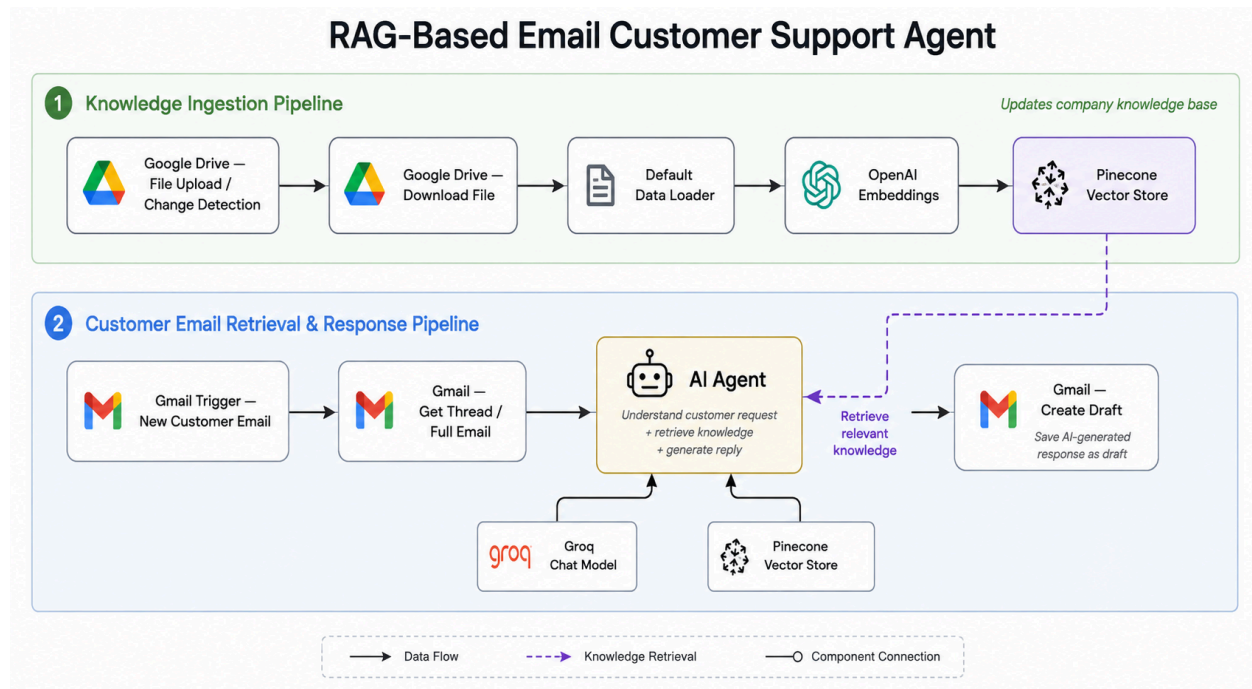


System Workflow



Ingestion Pipeline : In the google drive company PRD is uploaded and then sent to pinecone vector DB for embedding and chunking . Pinecone uses Open AI embedding and Default loader for chunking . Here the pdf is divided in to different chunks as vectors which is than used by retrieval pipeline for customer support

Knowledge Pipeline : When a user sends an email asking query the email body is sent to AI agent where The agent uses the embedded data from the Vector stores and provided relevant answers to write an email response

Tools Used : N8N , Gmail API , Open AI API , Pinecone

1. How the AI Agent Retrieves Company Information

The AI Agent receives the customer's email and identifies the information needed to answer it. It searches the Pinecone Vector Store using semantic similarity and retrieves the most relevant company documents. The retrieved information is then provided to the AI Agent as context.

2. How the System Generates an AI-Powered Reply

The AI Agent combines the customer's email with the relevant information retrieved from Pinecone. The Groq Chat Model then generates a clear, professional, and context-aware response based on the company's knowledge base.

3. How the System Sends the Response

The generated response is passed to the Gmail Create Draft node and saved as a draft in the support team's Gmail account. A human can review and edit the response before sending it to the customer.

4. Real-World Use Cases

- **E-commerce:** Answer questions about orders, refunds, shipping, and product policies.
- **SaaS Support:** Handle questions about pricing plans, features, API limits, and account policies.
- **Logistics:** Answer questions about shipments, warehouse policies, courier integrations, and service plans.

5. Limitations and Challenges

- **Knowledge Accuracy:** The AI can provide incorrect answers if the knowledge base is outdated or incomplete.
- **Retrieval Quality:** Poor document chunking or irrelevant search results can lead to inaccurate responses.
- **Complex Requests:** Some customer issues require account access, human judgment, or external systems that the AI agent cannot access.

6. Human-in-the-Loop

The system uses a human-in-the-loop approach by creating the AI-generated response as a Gmail draft instead of automatically sending it. A support representative reviews the draft, makes any necessary corrections, and then sends the final response to the customer. This provides an additional layer of quality control and reduces the risk of incorrect or inappropriate replies.